WORKED SOLUTION

STEP III 2006, Question 1

Pure • Curve Sketching

STEP III 2006, Question 1

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/06-s3-q1

The key idea

Divide out first: writing the curve as a straight line plus a small correction gives the oblique asymptote, tells you which side of it the curve lies, and makes the derivative easy. The derivative turns out to exceed 2 everywhere, and that single fact does most of the work in the root-counting parts: each equation rearranges to "curve = something simple", and we compare slopes.

Solution

The curve

Since $2x^3 - 10x = 2x(x^2 - 4) - 2x$,

$$y = \frac{2x(x^2-5)}{x^2-4} = 2x - \frac{2x}{x^2-4}.$$

Some features we can read off straight away:

- y is an *odd* function, so the curve has rotational symmetry of order 2 about the origin.
- It crosses the x-axis at $x = 0$ and $x = \pm\sqrt{5}$.
- **Asymptotes.** The denominator vanishes at $x = \pm 2$ (where the numerator is $\pm 4 \cdot (-1) \cdot 2 \neq 0$), so $x = 2$ and $x = -2$ are vertical asymptotes. As $|x| \to \infty$ the correction $2x/(x^2 - 4) \to 0$, so $y = 2x$ is an oblique asymptote.
- **Side of the asymptote.** $y - 2x = -2x/(x^2 - 4)$. For $x > 2$ this is negative (curve just below $y = 2x$); for $x < -2$ it is positive (just above). For $0 < x < 2$ it is positive and for $-2 < x < 0$ negative.
- **Behaviour near $x = \pm 2$.** As $x \to 2^+$, $-2x/(x^2 - 4) \to -\infty$, and as $x \to 2^-$ it tends to $+\infty$; by oddness, $y \to +\infty$ as $x \to -2^-$ and $y \to -\infty$ as $x \to -2^+$.

Differentiating the divided form,

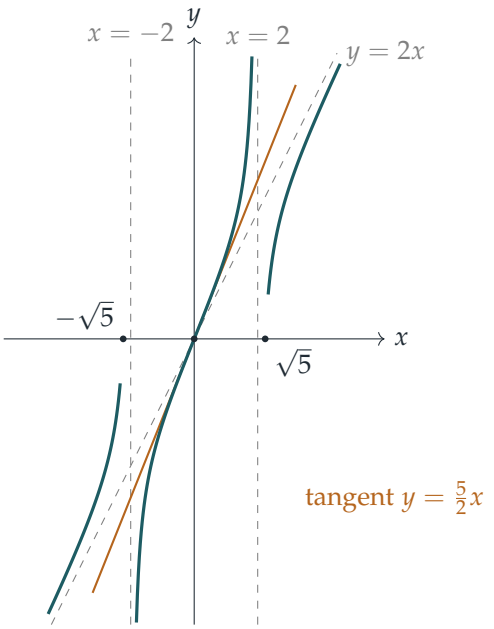
$$\frac{\mathrm{d}y}{\mathrm{d}x} = 2 - \frac{2(x^2-4) - 2x\cdot 2x}{(x^2-4)^2} = 2 + \frac{2(x^2+4)}{(x^2-4)^2}.$$

So $\mathrm{d}y/\mathrm{d}x > 2$ for every $x \neq \pm 2$: there are no stationary points, and on each of the three intervals $x < -2$, $-2 < x < 2$, $x > 2$ the curve is strictly increasing, rising from $-\infty$ to $+\infty$.

Tangent at the origin. At $x = 0$ the gradient is $2 + 8/16 = \frac{5}{2}$, so the tangent is $y = \frac{5}{2}x$. Differentiating once more,

$$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = -\frac{4x(x^2+12)}{(x^2-4)^3},$$

which vanishes only at $x = 0$ and changes sign there: the origin is a point of inflection. On $0 < x < 2$ the second derivative is positive (numerator and denominator both negative), so the middle branch is convex there and lies *above* its tangent $y = \frac{5}{2}x$; on $-2 < x < 0$ it is concave and lies below it.



Summary. Asymptotes $x = 2$, $x = -2$ and $y = 2x$; tangent at the origin $y = \frac{5}{2}x$.

In each part below, $x = \pm 2$ is never a root (the left-hand side is non-zero there while the right-hand side is zero), so we may divide by $x^2 - 4$ and count intersections of the curve with another graph.

Part (i)

Dividing by $\frac{3}{2}(x^2-4)$, the equation becomes

$$\frac{2x(x^2-5)}{x^2-4} = \tfrac{2}{3}(x+3),$$

so we want the intersections of the curve with the line $y = \frac{2}{3}x + 2$, of gradient $\frac{2}{3}$.

Ask yourself Why does a line of gradient less than 2 meet each branch exactly once?

Let $h(x)$ be the curve's y minus the line's. On each branch $h'(x) > 2 - \frac{2}{3} > 0$, so h is strictly increasing and has at most one zero. At the vertical asymptote ends $h \to \pm\infty$ just as y does (the line stays finite). At the outer ends, $h \approx 2x - \frac{2}{3}x = \frac{4}{3}x$, which tends to $-\infty$ as $x \to -\infty$ and to $+\infty$ as $x \to +\infty$. So on each of the three branches h runs from $-\infty$ to $+\infty$ and, by the intermediate value theorem, vanishes exactly once.

Hence equation (i) has **three** real roots. (Indeed it is the cubic $2x^3 - 3x^2 - 11x + 12 = 0$, with $x = 1$ as one root.)

Part (ii)

Dividing by $2(x^2 - 4)$, we need the intersections with $y = \frac{5}{2}x - 1$: the tangent at the origin moved down by 1. Its gradient now exceeds 2, so the argument of (i) no longer applies, and we look branch by branch.

- $x > 2$: the curve lies below $y = 2x$, while the line satisfies $\frac{5}{2}x - 1 - 2x = \frac{1}{2}x - 1 > 0$, so it lies above $y = 2x$. No intersection.
- $x < -2$: the curve lies above $y = 2x$, and the line lies below it since $\frac{1}{2}x - 1 < 0$. No intersection.
- $0 \leq x < 2$: the curve lies on or above its tangent $y = \frac{5}{2}x$ (convexity), which is above the line. No intersection.
- $-2 < x < 0$: as $x \to -2^+$ the curve tends to $-\infty$ while the line tends to -6; at $x = 0$ the curve is at 0, above the line's -1. So there is at least one crossing. Because $y'' < 0$ here, y' is decreasing on $(-2, 0)$, so $y' > y'(0) = \frac{5}{2}$; the difference "curve minus line" is strictly increasing, and the crossing is unique.

Hence equation (ii) has exactly **one** real root, lying between -2 and 0. (As a cubic it is $x^3 - 2x^2 + 8 = 0$, whose stationary values 8 and $\frac{184}{27}$ are both positive, confirming a single real root, near $x \approx -1.51$.)

Part (iii)

Dividing by $(x^2 - 4)^2$ gives

$$\left(\frac{2x(x^2 - 5)}{x^2 - 4}\right)^2 = x^2 + 1,$$

so we need the intersections of the curve with the rectangular hyperbola $y^2 - x^2 = 1$, that is, with $y = \sqrt{x^2 + 1}$ (upper branch) or $y = -\sqrt{x^2 + 1}$ (lower branch).

Ask yourself Can we use the symmetry to halve the work?

Yes. The map $(x, y) \mapsto (-x, -y)$ sends the curve to itself (it is odd) and swaps the two branches of the hyperbola. So intersections with the lower branch correspond one-to-one with intersections with the upper branch, and we need only count the latter.

Let $u(x) = \sqrt{x^2+1}$. Then $|u'(x)| = |x|/\sqrt{x^2+1} < 1$, so the difference $y - u$ has derivative greater than $2 - 1 = 1 > 0$ on each branch of the curve, and each branch gives at most one intersection. Now check the ends:

- $-2 < x < 2$: $y - u$ runs from $-\infty$ to $+\infty$. One root.
- $x > 2$: $y - u \to -\infty$ as $x \to 2^+$, and $y - u \approx 2x - x \to +\infty$ as $x \to \infty$. One root.
- $x < -2$: $y - u \approx 2x - |x| = 3x \to -\infty$ as $x \to -\infty$, and $y - u \to +\infty$ as $x \to -2^-$. One root.

So the curve meets the upper branch three times and, by symmetry, the lower branch three times. The two branches have no points in common ($u \geq 1$), so these six points have six distinct x-values, and equation (iii) has **six** real roots. Since it is a polynomial equation of degree 6 (namely $3x^6 - 33x^4 + 92x^2 - 16 = 0$), all its roots are real.

Checks

- Near $x = 0$, $y \approx 2x(-5)/(-4) = \frac{5}{2}x$, agreeing with the tangent.
- At $x = 1000$, $y - 2x \approx -0.002$: small and negative, as predicted.
- A numerical sign-change search on the original equations finds roots $x \approx -2.212,\ 1,\ 2.713$ for (i); only $x \approx -1.510$ for (ii); and $x \approx \pm 0.431,\ \pm 2.155,\ \pm 2.484$ for (iii).

Common slips

- Forgetting to check that $x = \pm 2$ are not roots before dividing by $x^2 - 4$.
- In (iii), counting only the intersections with $y = \sqrt{x^2+1}$ and missing the negative square root.
- In (ii), assuming "steep line, so one root per branch" as in (i): with gradient above 2 the outer branches need a separate argument.

Beyond STEP

The root counting here is the intermediate value theorem combined with monotonicity: a continuous, strictly increasing function that goes from $-\infty$ to $+\infty$ has exactly one zero. In a first-year analysis course both facts are proved carefully from the completeness of the real numbers, and the same reasoning underpins results such as "every real polynomial of odd degree has a real root". Abbott's *Understanding Analysis* develops this beautifully.

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